

5(a)	When the switch is opened, the reading is zero. When the switch is closed, the reading is non-zero for a short time. When the switch is reopened, the reading change sign and is non-zero for a short time. But the reading lasts for a short time and becomes zero again.
5(b)	When the switch is closed, current flows through the primary coil and create a magnetic field. Magnetic flux lines are linked from the primary coil to the secondary coil by the core. There is an increase in magnetic flux linkage with time in the secondary coil. By Faraday's law, e.m.f is induced in the secondary coil. When the switch is reopened, no current flows through the primary coil and hence no magnetic field. There is a decrease in magnetic flux linkage with time in the secondary coil. By Faraday's law, e.m.f of opposite polarity is induced in the secondary coil. But the reading becomes zero again after the switch is opened. No current flows and hence there is no magnetic field and there is no change of magnetic flux linkage with time in the secondary coil. Hence, no e.m.f induced.